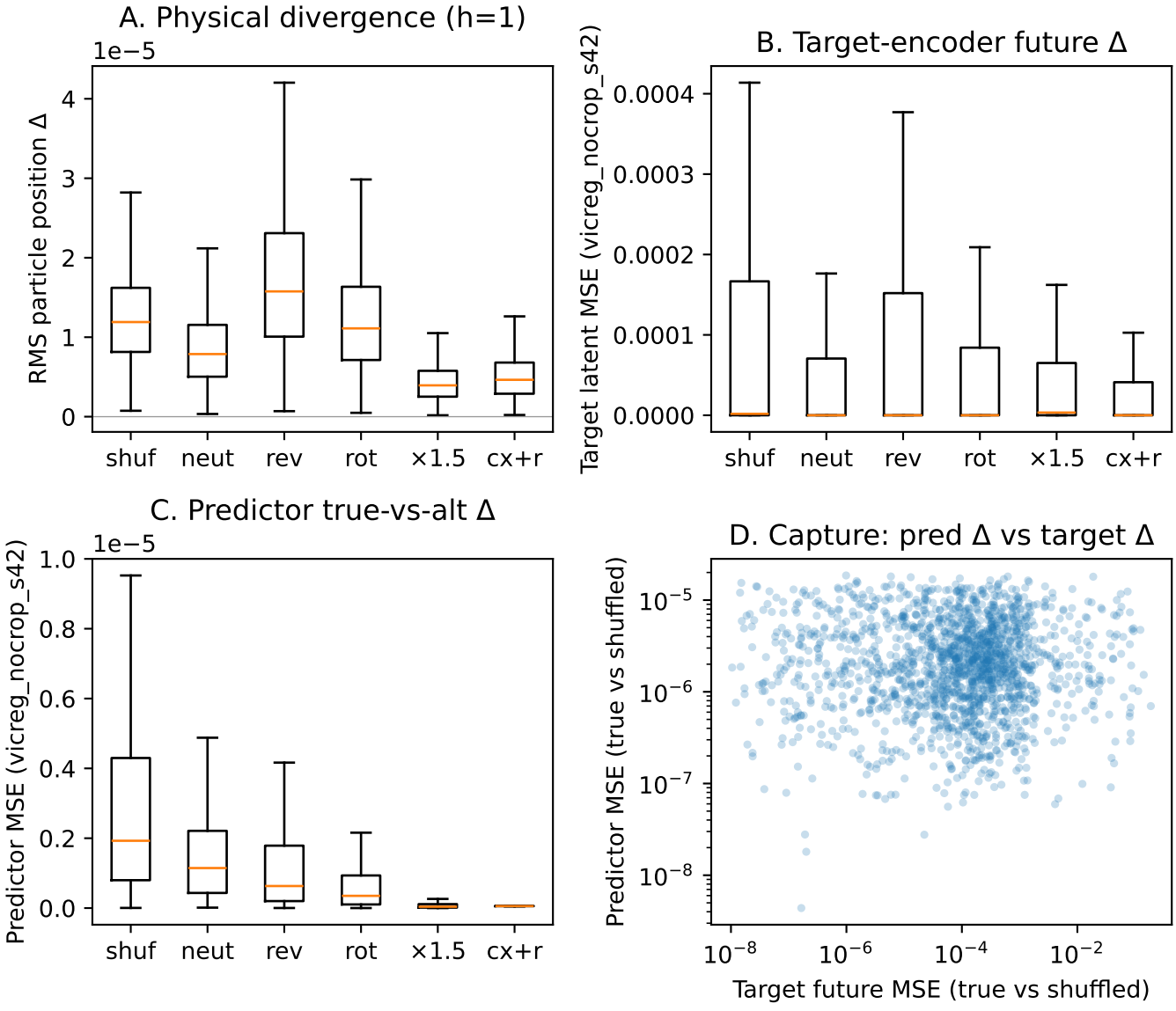
A. Physical divergence (h=1)
1e−5
RMS particle position Δ
0
1
2
3
4
shuf
neut
rev
rot
×1.5
cx+r
B. Target-encoder future Δ
Target latent MSE (vicreg_nocrop_s42)
0.0000
0.0001
0.0002
0.0003
0.0004
shuf
neut
rev
rot
×1.5
cx+r
C. Predictor true-vs-alt Δ
1e−5
Predictor MSE (vicreg_nocrop_s42)
0.0
0.2
0.4
0.6
0.8
1.0
shuf
neut
rev
rot
×1.5
cx+r
D. Capture: pred Δ vs target Δ
Predictor MSE (true vs shuffled)
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-8}
10^{-6}
10^{-4}
10^{-2}
Target future MSE (true vs shuffled)